



QUALITY MANAGEMENT BRANCH

STANDARD OPERATING PROCEDURES

FOR

VERIFICATION OF
HIGH FLOW TRANSFER STANDARDS

Standards Laboratory SOP 002

Revision 1

MONITORING AND LABORATORY DIVISION

JULY 2014



Approval of Standard Operating Procedures (SOP)

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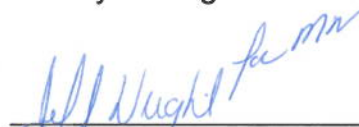
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1.0 Introduction

This standard operating procedure (SOP) describes the certification process of high flow transfer standards with respect to precision and accuracy. The Standards Laboratory certifies high volume (HiVol) orifices (flow rate of 40 cubic feet per minute) for use as high flow transfer standards against a high flow primary standard. High flow transfer standards are used to calibrate the flow rate of particulate matter samplers in the field. The performance of a HiVol air sampler is determined by the accurate calibration of flow reference standards (orifices).

2.0 Summary of Method

The calibration process for guest orifices is performed by measuring five points across the range of the orifice by drawing a high volume of air through a precision flow device (roots meter). Each sampling period must exceed three minutes. The pressure differential is recorded between the guest orifice and roots meter using manometers. Flow rates are adjusted by slowly closing the air inlet valve on a variable orifice or reducing the transformer voltage for the fixed and digital orifices. The results are then entered into an Excel spreadsheet which calculates the calibration formulas.

3.0 Summary of Changes to Previous Version

- Revision 1 introduces a new numerical sequence for this SOP and a number of formatting changes to comply with new SOP requirements.
- The Elgar, AC Power Source, Model 1001SL, was added to eliminate commercial AC power fluctuations that were adversely affecting the stability of the high volume sample motor.
- Electronic manometers were also incorporated; their digital readouts confirm the readings taken from the water manometers.
- Procedures were developed to certify digital fixed orifices.

4.0 Definitions

- Calibration - Establishes a correction formula to be used to adjust or correct the display of the guest instrument. This is determined through a comparison of the guest instrument to a known, or reference standard.
- Certification – Establishes traceability of a transfer standard to the roots meter. The certification of an instrument requires the results of two calibrations to meet requirements established by U.S. EPA.

- Recertification – Establishes traceability of a transfer standard to the roots meter using the current, one-time calibration along with previously completed calibration. Recertification requirements are the same as Certification.
- Flowmeter - An instrument for measuring the rate of flow of a fluid moving through a pipe or duct systems. The instrument is calibrated to give volume or mass rate of flow.
- HiVol Air Sampler - A device for sampling large volumes of an atmosphere for collecting the contained particulate matter by filtration. A HiVol consists of a high-capacity blower, a filter to collect suspended particles, and a means for measuring the flow rate. The PM10 Volumetric Flow Controller Sampler draws air into a specifically shaped inlet at 40 cubic feet per minute.
- Mass Flowmeter - A device that measures the mass flow rate of air passing a point, usually using the rate of cooling or heat transfer from a heated probe.
- Primary Flow Rate Standard - A device or means of measuring flow rate based on direct primary observations such as time and physical dimensions.
- Standard - A concept that has been established by authority, custom, or agreement to serve as a model or rule in the measurement of quantity of the establishment of a practice or procedure.
- Traceability to NIST - Documented procedure by which a standard is related to a more reliable standard verified by the National Institute of Standards Technology (NIST).
- Verification – Establishes a comparability of a candidate instrument to a known, or reference standard. The output of the candidate instrument is not corrected based upon the results of the verification procedures.

5.0 Personnel Qualifications

A thorough understanding of the high flow calibration process is essential. Before new personnel perform this procedure, one or more weeks of training from Standards Laboratory staff is required. They must also demonstrate competency in performing this procedure without assistance.

6.0 Health and Safety

The use of personal hearing protection is strongly recommended even though the operation of the HiVol test stand does not reach the noise exposure time of 85 decibels for a maximum limit of eight hours per day.

7.0 Cautions

- 7.1 Never perform the leak check procedure (Section 12.1) with a manometer connected to the venturi port on the guest orifice or the Manometer Isolation valve (Figure 12) open. Liquid from the manometer could be drawn into the system and cause motor damage.
- 7.2 While performing the leak check procedure (Section 12.6), do not block the air intake for more than 30 seconds. The high volume air motor could be damaged.

8.0 Interferences

Commercial AC power must be extremely stable since any fluctuation in power could adversely alter the actual air flow through the guest orifice.

9.0 Equipment and Supplies

- Elgar, AC Power Source, model 1001SL
- Dresser Roots Meter, model 5M175, counter with electronic transmitter used as a primary standard
- KEP MINItrol series electronic counter
- HiVol sample air motor
- Variable autotransformer (VARIAC)
- Water manometer (0-50 inch H₂O)
- Water manometer (0-16 inch H₂O)
- Digital manometer (0-50 inch H₂O)
- Digital manometer (0-10/40 inch H₂O)
- NIST certified digital thermometer
- NIST certified decimal stopwatch
- Devices to block air flow during leak checks
- Assorted sizes of tubing and plugs

10.0 Instrument Siting Requirements

A maintenance/laboratory environment suitable for electronic equipment operation.

11.0 Test Bench Setup Procedures

- 11.1 Figure 1 is a pictorial view of the HiVol test bench with all its components. Figure 2 illustrates the schematic configuration.



Figure 1: High Volume Flow Test Bench

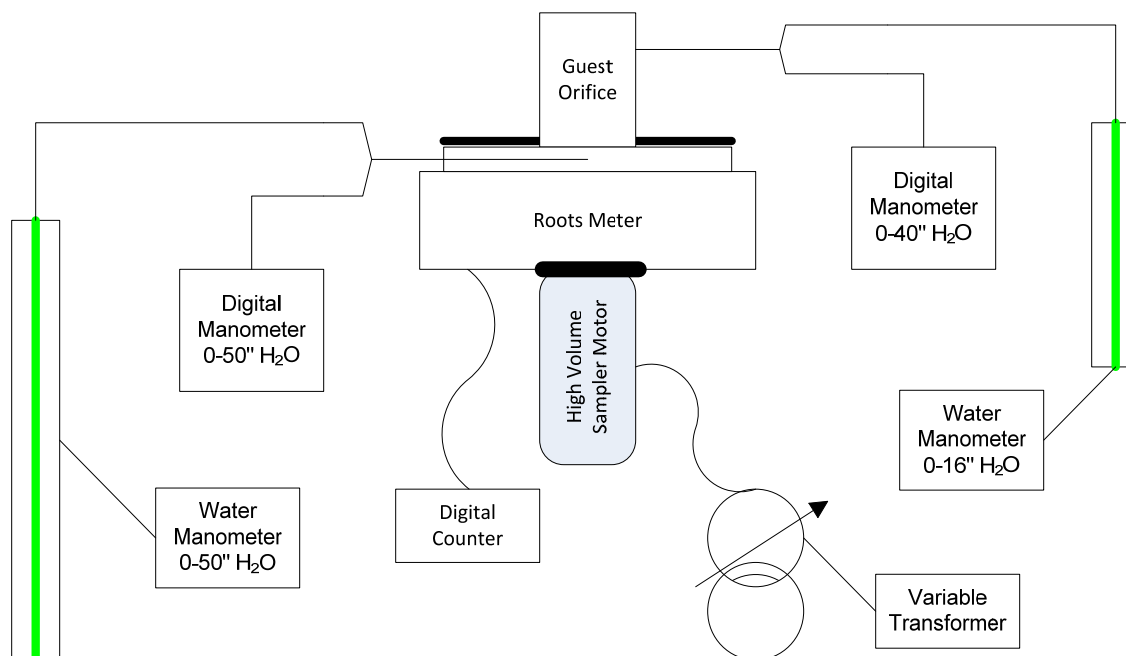


Figure 2: High Volume Flow Test Bench Schematic Diagram

- 11.2 Verify the VARIAC transformer, shown in Figure 3, is switched off and dialed counterclockwise to a low power setting. The power switch for the VARIAC is located on the outlet side of the unit.



Figure 3: VARIAC Transformer

- 11.3 To setup the Elgar AC Power Source, ensure the “Freq. Range” switch is set to “X1,” and the frequency dials are set to “60.00.” Switch on the power source and confirm the “AC Volts” meter is displaying 110 volts AC. If adjustment is necessary, use the “Amplitude” potentiometer. The dial settings are shown in Figure 4.

Note: The parameters on the Elgar AC Power Source seldom need adjustment.



Figure 4: Elgar AC Power Source

- 11.4 Zero the 50-inch water manometer by adding or removing green dyed water concentrate from the water reservoir so that the bottom of the meniscus is at the zero marker as shown in Figure 5.



Figure 5: Zero Level of the 50-inch Manometer

- 11.5 Align the 16-inch Slack Tube Water Manometer's sliding zero adjustment to the water meniscus as shown in Figure 6. To replenish the indicator fluid, add green dyed water concentrate to the top of the water column until the fluid level is at the middle point.



Figure 6: Zero Alignment of the 16-inch Manometer

- 11.6 Verify the stopwatch is displaying the proper mode. Locate and press the mode button until the "CHRONO" mode is displayed. Then press the "LAP/SPLIT•RESET" button until 1/1000m is selected.
- 11.7 On the guest's HiVol transfer standard undergoing certification, check the neoprene gasket for tears and/or foreign material. An airtight seal is essential to pass the leak test.
- 11.8 The HiVol transfer standard can be broken down into two parts, which are the orifice and the base plate. Verify the orifice is tightened sufficiently against the base plate to prevent leaks.

- 11.9 A fixed HiVol transfer standard consists of a single air intake hole of specific diameter as shown in Figure 7.



Figure 7: Fixed HiVol Orifice

- 11.10 A variable HiVol transfer standard consists of four air intake holes and an adjustment knob. Rotating the knob controls the airflow through the orifice. An example is shown in Figure 8.



Figure 8: Variable HiVol Orifice

11.11 A digital, fixed HiVol transfer standard is shown in Figure 9.



Figure 9: Digital Fixed Orifice

- 11.12 Place the guest HiVol transfer standard on the Roots Meter interface assembly so that the differential pressure gauge (if attached) is visible to the user and secure using four wing nut screws as shown in Figure 10.



Figure 10: Securing a Flow Standard

Note: Some clients deliver their HiVol transfer standards without a gauge.

- 11.13 If the HiVol transfer standard does not have a differential pressure gauge, use the laboratory's digital manometer and the 0-16 inch water manometer.
- 11.14 Attach the tubing from the 0-16 inch water manometer to the venturi port on the guest's high flow transfer standard. The venturi port is shown in Figure 11.



Figure 11: Venturi Port

12.0 LEAK CHECK PROCEDURES

- 12.1 Before performing the leak check, close the valve connecting the Roots Meter's digital and 0 – 50 inch water manometer which is shown in Figure 12.

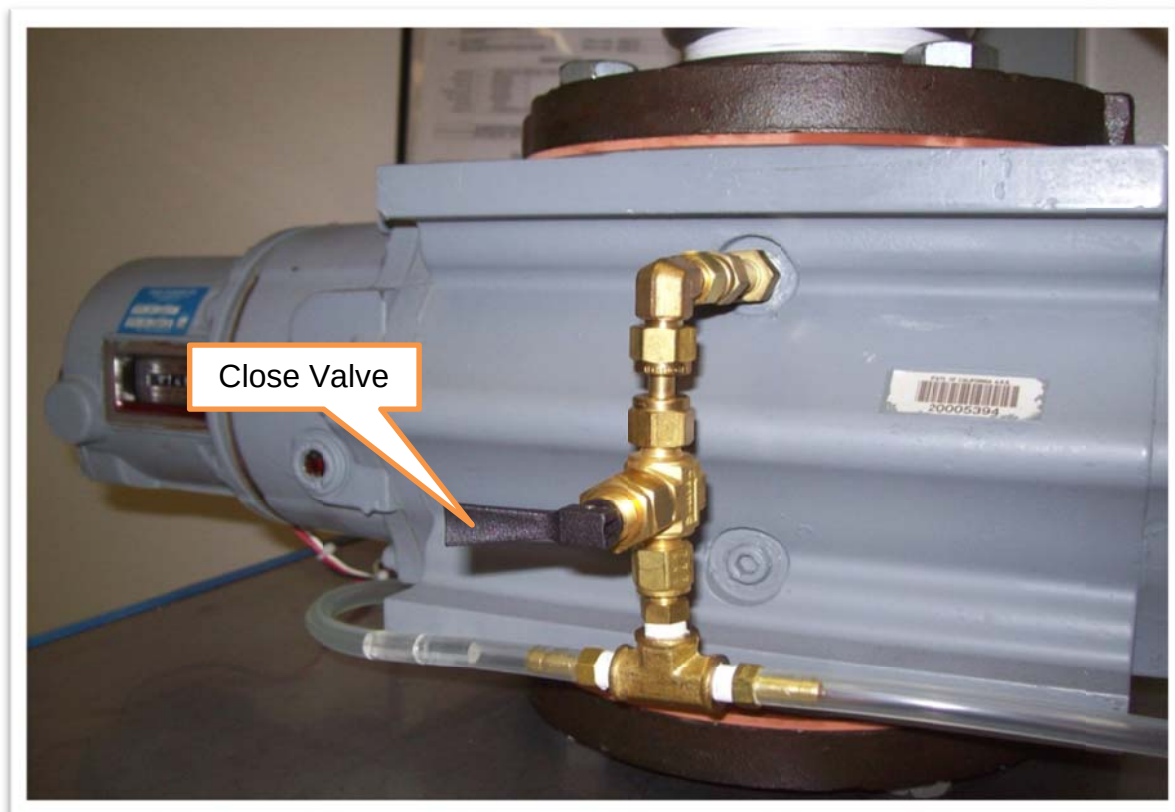


Figure 12: Manometer Isolation Valve

- 12.2 If the HiVol transfer standard comes with a differential pressure-measuring device, disconnect it from the venturi port as shown in Figure 11.
- 12.3 Attach a plug to the venturi port of the HiVol transfer standard.
- 12.4 Turn on the VARIAC and rotate the black dial clockwise until the display reads approximately 50 volts AC.

- 12.5 Follow the specific instructions below to perform a leak test on the fixed and variable orifices.
- Fixed orifice – Block the airflow through the guest HiVol transfer standard by placing a small rubber mat over the air intake hole. Refer to Figure 7.
 - Variable orifice – Turn the flow adjustment knob on the guest HiVol transfer standard for maximum flow (counter clockwise). Block all four air intake holes by inseting rubber stoppers into the openings. Refer to Figure 8.
 - Digital orifice – A leak check cannot be performed on the digital orifice. Refer to Figure 9.
- 12.6 Once the air flow is blocked, immediately start the stopwatch while monitoring the roots meter counter and record the starting number from the counter on the Analog HiVol Orifice Certification Data Sheet, Figure 14, in the Leak Check section, under “Initial.” After 20 seconds, or 0.33 as displayed on the decimal stop watch in the 1/1000M mode, record the counter number in the block labeled “Final.” The tolerance for the leak check must be less than 1 cubic foot per minute.
- CAUTION:** Once the final reading is recorded, immediately remove the flow blocking device. Do not block the air intake for more than 30 seconds or damage may occur to the HiVol sample air motor.
- 12.7 A leak is indicated if the tolerance is greater than 1 cubic foot per minute. The leak must be repaired before continuing the test. The severity and location of the leak will determine if repair can be accomplished on the test stand or whether it must be returned to the client.
- 12.8 Open the roots meter’s manometer isolation valve, Figure 12, and connect the orifice manometer to the venturi port, Figure 11.

13.0 CALIBRATION PROCEDURE

- 13.1 Enter the guest instrument information into the HiVol Log as shown in Figure 13. The log is located at: "W:\Standards Laboratory\Equipment Logs\Logs.mdb"

HiVol Log										
Date	Log No	Property N	Action	Agency	Type	Init	Re-Run	Pass	Remarks	
11/5/2013	2013-317	20112275	Re-Cert	Quality Assurance	Fixed-Digital	JF	☐	☒		
11/5/2013	2013-318	20121445	Full-Cert	Quality Assurance	Fixed-Digital	JF	☒	☐		
11/5/2013	2013-319	20121447	Full-Cert	Quality Assurance	Fixed-Digital	JF	☒	☐		
11/5/2013	2013-320	20121446	Full-Cert	Quality Assurance	Fixed-Digital	JF	☒	☐		
11/5/2013	2013-321	107644	Re-Cert	Quality Assurance, Van C	Variable	JF	☐	☒		
11/6/2013	2013-318	20121445	Full-Cert	Quality Assurance	Fixed-Digital	JF	☒	☐		
11/6/2013	2013-319	20121447	Full-Cert	Quality Assurance	Fixed-Digital	JF	☐	☒		
11/6/2013	2013-320	20121446	Full-Cert	Quality Assurance	Fixed-Digital	JF	☐	☒		
11/7/2013	2013-320	107643	Re-Cert	Quality Assurance, Van X	Variable	JF	☐	☒		
11/7/2013	2013-318	20121445	Full-Cert	Quality Assurance	Fixed-Digital	JF	☐	☒		
11/7/2013	2013-328	107643	Re-Cert	Quality Assurance, Van X	Variable	JF	☐	☒		
11/21/2013	2013-333	20005450	Re-Cert	AM - South	Variable	JF	☐	☒		
1/10/2014	2014-007	20003935	Re-Cert	Standards Lab	Fixed	JF	☐	☒	1st Qrt Cert	
1/17/2014	2014-016	107338	Re-Cert	San Diego County APCD	Variable	JF	☐	☒		
1/17/2014	2014-017	106011	Re-Cert	San Diego County APCD	Variable	JF	☐	☒		
1/21/2014	2014-018	106012	Re-Cert	San Diego County APCD	Variable	JF	☐	☒		
1/30/2014	2014-031	20004049	Re-Cert	Special Purpose Monitorin	Fixed	JF	☐	☒		
2/14/2014	2014-060	106060	Full-Cert	San Diego APCD	Fixed-Digital	JF	☒	☐		
2/14/2014	2014-062	107375	Re-Cert	AM - North	Variable	JF	☐	☒		
2/14/2014	2014-065	20071203	Full-Cert	AM - South	Variable	JF	☒	☐		
2/18/2014	2014-060	106060	Full-Cert	San Diego APCD	Fixed-Digital	JF	☐	☒		
2/18/2014	2014-065	20071203	Full-Cert	AM - South	Variable	JF	☐	☒		
2/25/2014	2014-067	105839	Re-Cert	AM - North	Variable	JF	☐	☒		
2/26/2014	2014-068	20002031	Re-Cert	AM - North	Fixed	JF	☐	☒		
3/6/2014	2014-077	20005448	Re-Cert	AM - North	Variable	JF	☐	☒		
3/13/2014	2014-084	105868	Re-Cert	AM - North	Variable	JF	☐	☒		
3/13/2014	2014-085	20004090	Re-Cert	AM - North	Fixed	JF	☐	☒		
4/7/2014	2014-104	107309	Re-Cert	Shasta County AQMD	Variable	JF	☐	☒		
4/10/2014	2014-111	20003935	Re-Cert	Standards Lab	Fixed	JF	☐	☒	2nd Qrt Cert/Pre Cal	
4/10/2014	2014-114	20004255	Re-Cert	AM - North	Fixed	JF	☐	☒		
4/17/2014	2014-128	107324	Re-Cert	Bay Area AQMD	Variable	JF	☐	☒		

Figure 13: HiVol Log Entries

- 13.2 Record the following information on the appropriate data sheet: "Analog High Volume Orifice Certification Data Sheet," Figure 14, or the "Digital High Volume Orifice Certification Data Sheet," Figure 15.

- Log Number from the Standard Laboratory check-in log
- Property or Barcode Number for the guest standard
- Agency Name for the guest standard
- Name of staff performing the test
- Current Date
- ARB Roots Meter property number
- Ambient pressure in mmHg at the start of test
- Name of staff performing the quality assurance check
- Enter the Orifice and Roots Pressure readings from the previous calibration; this information is useful in quickly determining performance.

MLD STANDARDS LABORATORY
ANALOG HIVOL ORIFICE CERTIFICATION DATA SHEET

Log #: 2014 -
 Bar Code #: _____
 Agency: _____
 Calibrated by: Jerry

Date:

M	M	D	D	Y	Y
				1	4

 Roots Meter: 20005394
 Ambient Pressure mmHg: _____
 Checked by: _____

Roots Meter		Elapsed Time	Orifice Pressure ("H ₂ O)			Roots Pressure ("H ₂ O)			Ambient Temp (deg C)		
Final	Initial		Final	Initial	Average	Final	Initial	Average	Final	Initial	Average

Leak Check =

Final:	
Initial	
Difference	
Passing	< 1.0

Previous Cal

Orifice	Roots

Comments:

Figure 14: Analog HiVol Orifice Certification Data Sheet

MLD STANDARDS LABORATORY
DIGITAL HiVOL ORIFICE CERTIFICATION DATA SHEET

Log #: 2014 - _____
 Bar Code #: _____
 Agency: _____
 Calibrated by: Jerry

Date:

					1		4
--	--	--	--	--	---	--	---

 Roots Meter: 20005394
 Ambient Pressure mmHg: _____
 Checked by: _____

Roots Meter		Elapsed Time	Orifice Pressure ("H ₂ O)			Roots Pressure ("H ₂ O)			Ambient Temp (deg C)		
Final	Initial		Final	Initial	Average	Final	Initial	Average	Final	Initial	Average

Orifice	Roots

Digital HiVol QA			Digital HiVol QS		
Final	Initial	Average	Final	Initial	Average

Figure 15: Digital HiVol Orifice Certification Data Sheet

- 13.3 To prepare the Digital HiVol Orifice for certification, disassemble the case, and then remove the plug from the T connector as shown in Figure 16.

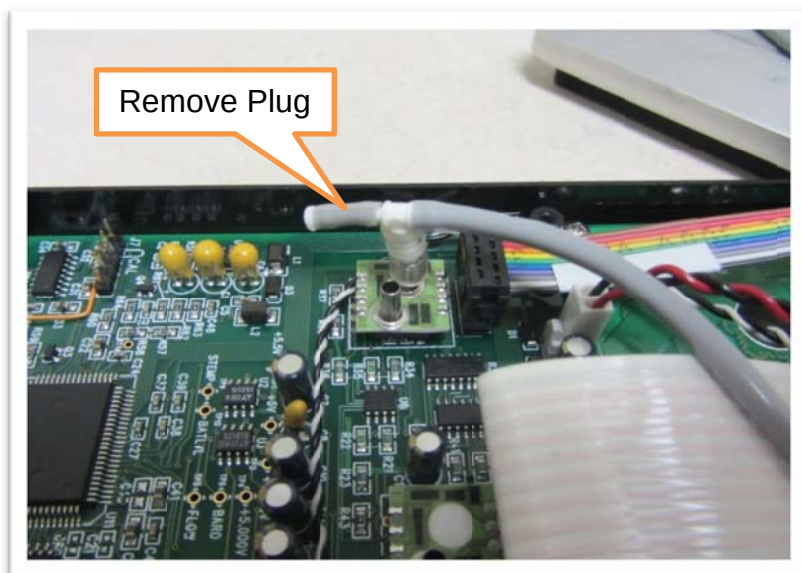


Figure 16: Factory Installed Plug

- 13.4 Install tubing that connects to the Venturi manometer as shown in Figure 17.

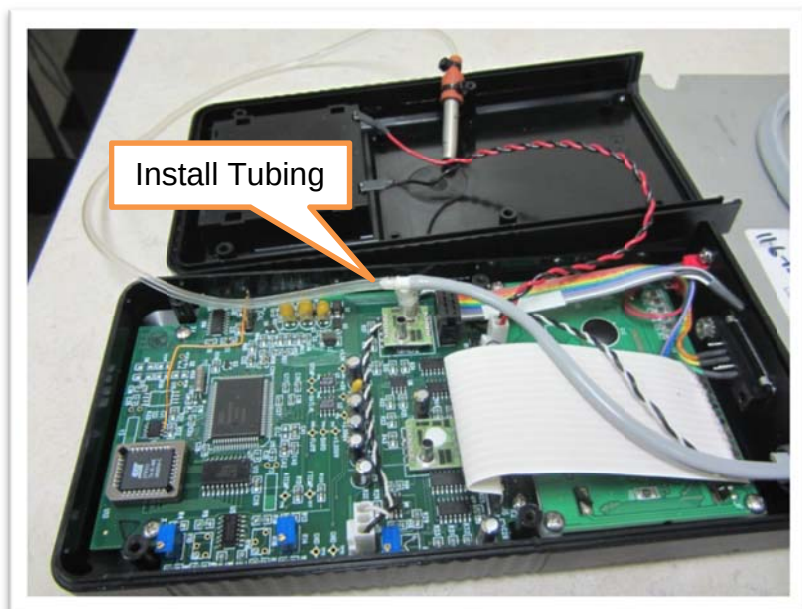


Figure 17: Tubing Installation

13.5 The following certification procedures are applicable for all HiVol transfer standards.

13.6 Adjust the VARIAC to the first (high) set point.

NOTE: Set points are determined by the range of the display devices provided with the guest orifices, which are usually a magnehelic gauge or a digital manometer. Set points can also be determined by referring to the guest HiVol's previous calibration folder, or, if new, the data from a similar orifice.

13.7 Allow the flow to stabilize and the high volume motor to warm-up for 5 minutes. The air flow through the Roots Meter is measured by the counter; the counter is controlled by the Counter Control Switch. The control panel is shown in Figure 18.

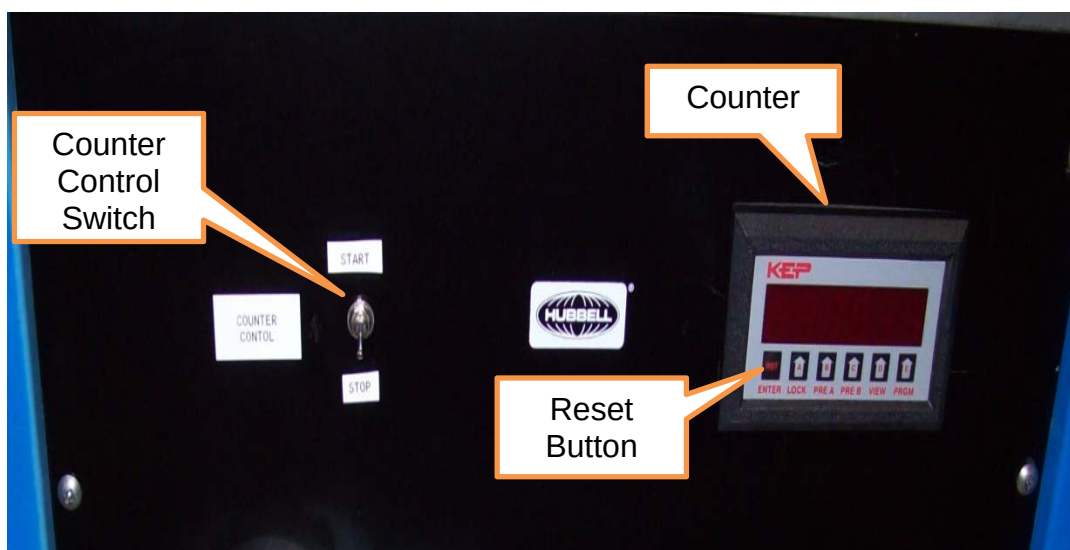


Figure 18: Counter Control Panel

13.8 As airflow passes through the Roots Meter, the counter will increment. Place the Counter Control Switch to "STOP." Reset the counter to zero. Once all the readings have stabilized, record the following information on the appropriate High Volume Orifice Certification Data Sheet that was prepared in Section 13.2.

- In the "Initial" block of the "Roots Meter" Section, record the number from the counter to the nearest tenth (the number displayed on the counter prior to switching to the "START" position).
- In the "Initial" block of the "Ambient Temp" Section, record the temperature reading displayed on the digital thermometer.
- In the "Initial" block of the "Roots Pressure" Section, record the pressure reading displayed on the digital manometer and verified against the 50 inch water manometer.
- In the "Initial" block of the "Orifice Pressure" Section, record the pressure reading displayed on the digital manometer and verified against the 16 inch water manometer.
- For a digital HiVol orifice, in the "Initial" block of the "Digital HiVol QA and QS" Sections, record the readings from the orifice's digital display.

13.9 Simultaneously, start the stopwatch and flip the Counter Control switch to "START."

13.10 Allow airflow to pass through the guest HiVol transfer standard and Roots Meter for a minimum of 3 minutes.

13.11 After 3 minutes, perform the following:

- Simultaneously, stop the stopwatch and flip the Counter Control switch to "STOP."
- In the "Final" block of the "Roots Meter" Section, record the number from the counter to the nearest tenth.
- In the "Elapsed Time" block, record the time from the stopwatch.
- In the "Final" block of the "Ambient Temp" Section, record the temperature reading displayed on the digital thermometer.
- In the "Final" block of the "Roots Pressure" Section, record the pressure reading displayed on the digital manometer and verified against the 50 inch water manometer.
- In the "Final" block of the "Orifice Pressure" Section, record the pressure reading on the digital manometer and verified against the 16 inch water manometer.
- For a digital HiVol orifice, in the "Final" block of the "Digital HiVol QA and QS" Sections, record the readings from the orifice's digital display.

13.12 Average the “Initial” and “Final” numbers for temperature and pressure collected in the previous steps. Place those numbers in the “Average” block.

13.13 Advance to the next calibration step by reducing the flow rates according to the following procedures:

- Fixed and Digital HiVol Transfer Standards: Adjust the VARIAC to select all flow rates.
- Variable HiVol Transfer Standards: Adjust the flow control knob (Figure 11). DO NOT adjust the VARIAC.

13.14 Repeat steps 13.8 through 13.13 until the remaining four flow measurements are recorded. DO NOT reset the counter to zero between each step.

13.15 When all the steps are completed, turn the black dial of the VARIAC counterclockwise and switch the power switch to the off position.

14.0 Data Acquisition and Storage

- 14.1 Follow the displayed data path and open the “Hi-Vol Cal Data.xlsm workbook” shown in Figure 19.

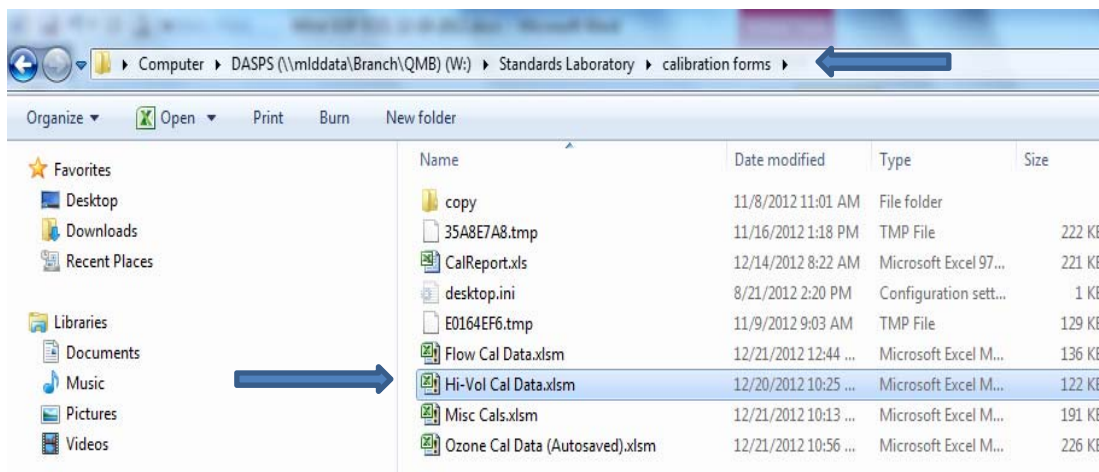


Figure 19: “Hi-Vol Cal Data.xlsm” workbook

- 14.2 Open “Login” worksheet. See Figure 20.
- 14.3 To initially log an instrument into the Standards Laboratory List, enter the “Instrument ID” number into cell G4. Then press “LOGIN”.
- 14.4 To enter calibration data, enter the “Instrument ID” number into cell G4. Then press “Collect Customer Info”.

STANDARDS LABORATORY PROJECT CHECK LIST			
Agency: SAN DIEGO COUNTY APCD		Customer ID 140	LOGIN (1)
Contact:		Instrument ID 106060	
Year & Log Number: 2014060		Prev. Log. No.: N/A	
Instrument: BGI HIVOL ORIFICE, Digital		SN: 91	
Bar Code: 106060		Date: 2/12/2014	
PHONE #: 858-586-2789		Collect Customer Info (3)	
Log Number: 2014060			
Select Service: Certification (C) Verification (V)			
Enter Leter selection: V			
OZONE:			
Calibration Date (s)			
1 _____ []	2 _____ []	3 _____ []	
4 _____ []	5 _____ []	6 _____ []	
FLows:			
Calibration Date (s)			
POS 1 _____ []	2 _____ []	3 _____ []	4 _____ []
POS 2 _____ []	2 _____ []	3 _____ []	4 _____ []
POS 3 _____ []	2 _____ []	3 _____ []	4 _____ []
POS 4 _____ []	2 _____ []	3 _____ []	4 _____ []
HIVOLL ORIFICE			
Calibration Date (s)			
1 _____ []	2 _____ []	3 _____ []	

Figure 20: Instrument Login

- 14.5 Enter the data collected on the "MLD Standards Laboratory High Volume Orifice Calibration Data Sheet" into the shaded areas on the form. Once the form is completed, print a copy for record. See Figure 21.

CALIFORNIA AIR RESOURCES BOARD STANDARDS LABORATORY									
HI-VOL CALIBRATION DATA									
Calibration Date:		2/18/2014		Log Number:		2014060			
Property Number:		106060		Previous Log Number:		2014060			
Bar Code Number:		106060							
Instrument Make & Model:		BGI HIVOL ORIFICE, Digital							
Property of:		SAN DIEGO COUNTY APCD							
Ambient Pressure (Pa):		764.8							

Roots		Elapsed Time t, (min)	Orifice Press. P,	Roots Meter Pr, (H2O) Ave.	Ambient Temp. Ta, (° C) Ave.	Air Flow ft³/min		Meas. vs CALC RD, (%)	PASS/FAIL
						Qa	Qstd		
Final	Initial								
163.4	0.0	3.011	8.79	7.84	24.6	53.23	53.64	0.08	PASS
290.7	163.4	3.011	5.37	4.77	24.4	41.79	42.13	-0.20	PASS
377.9	290.7	3.013	2.55	2.27	23.3	28.78	29.13	0.05	PASS
432.9	377.9	3.011	1.02	0.91	22.6	18.23	18.49	0.54	PASS
470.2	432.9	3.013	0.45	0.41	22.8	12.37	12.54	-0.56	PASS

Note: mmHg = 25.4 * inch H2O / 13.6
RD (Qa) = (Qa Calculated - Qa Measured) / Qa Measured
RD (Qstd) = RD (Qa)

Qa CFM	Qstd, CFM
52.68	53.2
41.27	41.69
28.63	29.07
18.23	18.53
12.34	12.54

CALIBRATION EQUATIONS

Volume (Δ Vol) = Roots meter [final - initial]
Actual Volume (Va) = Δ Vol * (Pa - 25.4 * Pr / 13.6) / Pa
Actual Air Flow (Qa) = Va / t
Calc. Air Flow (Qcalc) = 28.61 * (P * [Ta + 273] / Pa) ^ ½ +
Std Air Flow (Qstd) = Qa * (Pa / 760) * (298 / (Ta + 273))

FINAL EQUATIONS

Actual Air Flow (Qa) = 28.61 * (P * [Ta + 273] / Pa) ^ ½ + 0.363
Standard Air Flow (Qstd) = 11.22 * (P * Pa / [Ta + 273] / Pa) ^ ½ + 0.369

Digital Meter:

Actual Air Flow (Qa CFM)	0.98479	*	Display	+	0.222
Standard Air Flow Qstd CFM)	0.98672	*	Display	+	0.235

Calibrated by: _____ Checked by: _____

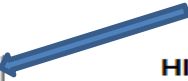
Figure 21: Hi-Vol Cal Data

NOTE: Criteria for Measured Flow vs. Calculated Flow Values:
Each of the 5 (Measured Flow vs. Calculated) flow measurements points must have a relative percent difference of ± 2%. If the criteria is not met, either maintenance or a full certification is required.

- 14.6 Open the “QASummary” worksheet and then click on the “Run” button. This will populate the previous calibration information. Once the form is completed, print a copy for record. See Figure 22.

STANDARDS LABORATORY

**HI-VOL TRANSFER STANDARD
CERTIFICATION SUMMARY SHEET**

Run 

Date: 2/18/2014
 Log Number: 2014060 Pre. Log Number: N/A
 Property Number: 106060
 Bar Code Number: 106060
 Instrument Make & Model: BGI HIVOL ORIFICE, Digital
 Property of: SAN DIEGO COUNTY APCD

Certification: X Re-Certification:

Comparison	Date	Slope	Intercept
1	2/14/2014	28.83	0.104
2	2/18/2014	28.61	0.363

Note: Air Flow (Qa) = Slope * (P * Ta (°K) / Pa (mmHg)) ^ ½ + Intercept

MEAN Slope = 28.72
 MEAN Intercept = 0.234
 STD. Dev. Slope = 0.1581
 STD. Dev. Intercept = 0.18332
 STD. Dev. Slope / Mean Slope = 0.00550 (< 0.007) **PASS**
 Full Scale [STD. Dev. Intercept / Max Range = 0.00344 (< 0.007) **PASS**

NEW CERTIFICATION RELATIONSHIP

Actual Air Flow (Qa) = 28.72 * (P * Ta (°K) / Pa (mmHg)) ^ ½ + 0.2336

Figure 22: QASummary “Run Button”

Note: Actual Flow-

- Relative Standard Deviation (RSD) of previous two slope values must be less than 0.007.
- Full scale Standard Deviation (FSRD) of previous two intercept values must be less than 0.007.
- If the criteria are not met, either maintenance or a full certification is required.

14.7 Open the "QstdSummary". Print a copy for record. See Figure 23.

HI-VOL TRANSFER STANDARD CERTIFICATION SUMMARY SHEET			
Date:	2/18/2014		
Log Number:	2014060		
Property Number:	106060		
Bar Code Number:	106060		
Instrument Make & Model:	BGI HIVOL ORIFICE, Digital		
Property of:	SAN DIEGO COUNTY APCD		
Certification:	X	Re-Certification:	

Comparison	Date	Slope	Intercept
1	2/14/2014	11.31	0.105
2	2/18/2014	11.22	0.369

Note: Air Flow (Qstd) = Slope * (P * Pa (mmHg) / Ta (°K)) ^ ½ + Intercept

MEAN Slope =	11.26
MEAN Intercept =	0.237
STD. Dev. Slope =	0.0665
STD. Dev. Intercept =	0.18699
STD. Dev. Slope / Mean Slope =	0.00590
Full Scale [STD. Dev. Intercept / Max Range =	0.00351

(< 0.007	PASS
(< 0.007	PASS

NEW CERTIFICATION RELATIONSHIP

Standard Air Flow (Qstd) = 11.26 * (P * Pa (mmHg) / Ta (°K)) ^ ½ + 0.237

Figure 23: QstdSummary

Note: Standard Flow-

- Relative Standard Deviation (RSD) Slope of 2 previous Slope values must be: less than 0.007.
- Full scale Standard Deviation (FSRD) Intercept of 2 previous intercept values must be: less than 0.007.
- If the criteria are not met, either maintenance or a full certification is required.

14.8 Open the "HiVol Cert Report". Print a copy for record. See Figure 24.

CALIFORNIA AIR RESOURCES BOARD HIVOL CERTIFICATION REPORT			
To:	SAN DIEGO COUNTY APCD	Log Number:	2014060
From:	Jerry Freeman Data Analysis and Special Projects Section	Calibration Date:	2/18/2014
		Report Date:	2/18/2014
IDENTIFICATION			
Instrument:	BGI HIVOL ORIFICE, Digital	Site Name:	MLD Standards Lab
Property No.:	106060	Site Number:	34-299
Serial No.:	106060	Location:	1927 13th Street Sacramento, CA 95811
Previous Log No.:	N/A		
Bar Code No.:	106060		
Elevation:	25.00'		
Property of:	SAN DIEGO COUNTY APCD		
CALIBRATION STANDARD		ID Number	
ROOTS METER - 5M175CEX		11361	
CALIBRATION RESULTS			
COMPONENT	Qa	Qstd	
Instrument Range	53	70	
Display of Delta P (Inches of Water)	8	8	
Previous Calibration Date	2/14/2014	2/14/2014	
Change From Previous Calibration	-0.78	-0.832	
Note: Ta = Ambient Temperature (k), Pa = Ambient Pressure (mmHg)			
		Certification Expires:	2/18/2015
Certification Equation:			
Actual Airflow (QA) =	28.72	$(\Delta P \cdot T_a / P_a)^{1/2} +$	0.234
Standard Airflow (Qstd) =	11.26	$(\Delta P \cdot P_a / T_a)^{1/2} +$	0.237
DIGITAL FLOW METER ONLY:			
Actual Air Flow (QA CFM)	0.9848	* Disq +	0.222
Standard Air Flow Qstd CFM)	0.9867	* Disq +	0.235
Calibrated by:	Checked by:		

Figure 24: HiVol Certification Report

14.9 For a guest standard that meets the required criteria, print two copies of the Verification/Calibration report. Place one copy with the client's High Flow transfer standard and file the other along with "CalReport" spreadsheet inside the guest folder. Create an electronic copy by scanning the High Flow Spreadsheet and Final report. Save as PDF files in the following location: W:\Standards Laboratory\Calibration Reports, under the Equipment Number.

14.10 Full Certification for guest high flow standards:

- Two tests run on different days.
- Each R2 must be 0.9999 or greater.
- All other re-certification criteria apply.

NOTE: Full certification service is required if the standard has been repaired, is new, has been factory calibrated since the last time it was in for testing, or fails to meet the re-certification criteria.

15.0 Quality Control

- 15.1 Leak checks are performed on the primary flow standard and the guest high flow transfer standard before each calibration.
- 15.2 A certification requires 2 consecutive assays each performed on different days. This ensures that each assay is performed at a different room ambient condition (temperature and barometric pressure) than the previous assay. This practice checks the stability of the guest flow instrument. Each assay must have a linear regression subjected to an R2 of greater than 0.9999.
- 15.3 A control check for the Roots Meter is performed quarterly using dedicated high volume orifice ARB #: 20003935, using both digital and water manometers. The results of the checks are entered into a control chart that tracks trends and displays control thresholds. The calibration requirement must be $\pm 2\%$ of the primary standard.
- 15.4 Biennially, the Roots Meter is sent to a vendor to perform NIST traceable certification. To verify the Roots Meter has not significantly changed following the certification, two operational assays will be performed; once prior to it being shipped to the vendor and again when it is returned to service.

NOTE: Prior to shipping the meter, drain both oil reservoirs. Before the meter is returned to service, fill both reservoirs with ROOTS Meter oil. The reservoirs are not sealed and will leak oil during shipment if not drained. Figure 25 displays the oil reservoir locations and proper oil level in the sight gauges.

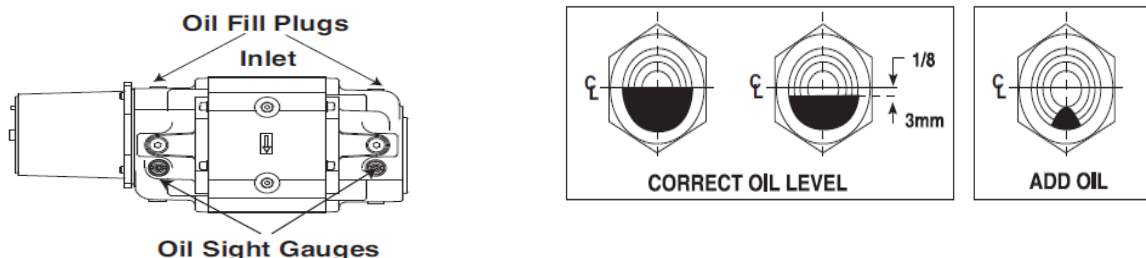


Figure 25: Roots Meter Oil Fill Plugs/Sight Gauges

16.0 Routine Service Checks

Before every assay, check the indicating fluid levels of the water manometers. If they require service, fill to the zero graduation on the scale with diluted green concentrate.

17.0 Preventive Maintenance and Repairs

Biennially, (while the Roots Meter is out for certification) check the carbon brushes in the high volume vacuum motor and replace if necessary.

NOTE: This is a low use motor and the brushes should last for a significant period of time.

18.0 Troubleshooting

- 18.1 If a HiVol orifice fails a leak check, ensure the orifice is securely attached to the orifice flange and the tubing connections are in good repair and snugly fastened. Ensure a gasket is properly installed between the orifice body and the mounting plate.
- 18.2 If an abnormally low or erratic indication is displayed on the manometers, ensure that all the tubing connections are tight and are servable. Replace components or tubing if necessary.

19.0 References

- Compendium Method IO-2.1, Sampling of Ambient Air for Total Suspended Particulate Matter (SPM) and PM10 using High Volume Sampler (EPA/625/R-96/010a)
- 40 CFR 50, Appendix B, Reference Method for the Determination of Suspended Particulate Matter in the Atmosphere (High - Volume Method)
- 40 CFR 50, Appendix J, Reference Method for the Determination of Suspended Particulate Matter in the Atmosphere
- Investigation of Flow Rate Calibration Procedures Associated with the High Volume Method for Determination of Suspended Particulates (EPA-600/4-78-047)
- U.S. EPA, QA Handbook for Air Pollution Measurement Systems, Volume II, Ambient Air Quality Monitoring Program. (EPA-454/B-08-003)

20.0 Data Calculations

All calculations are performed within the Excel HiVol Cal.xlsm workbook when the certification data is entered. (See Section 14)